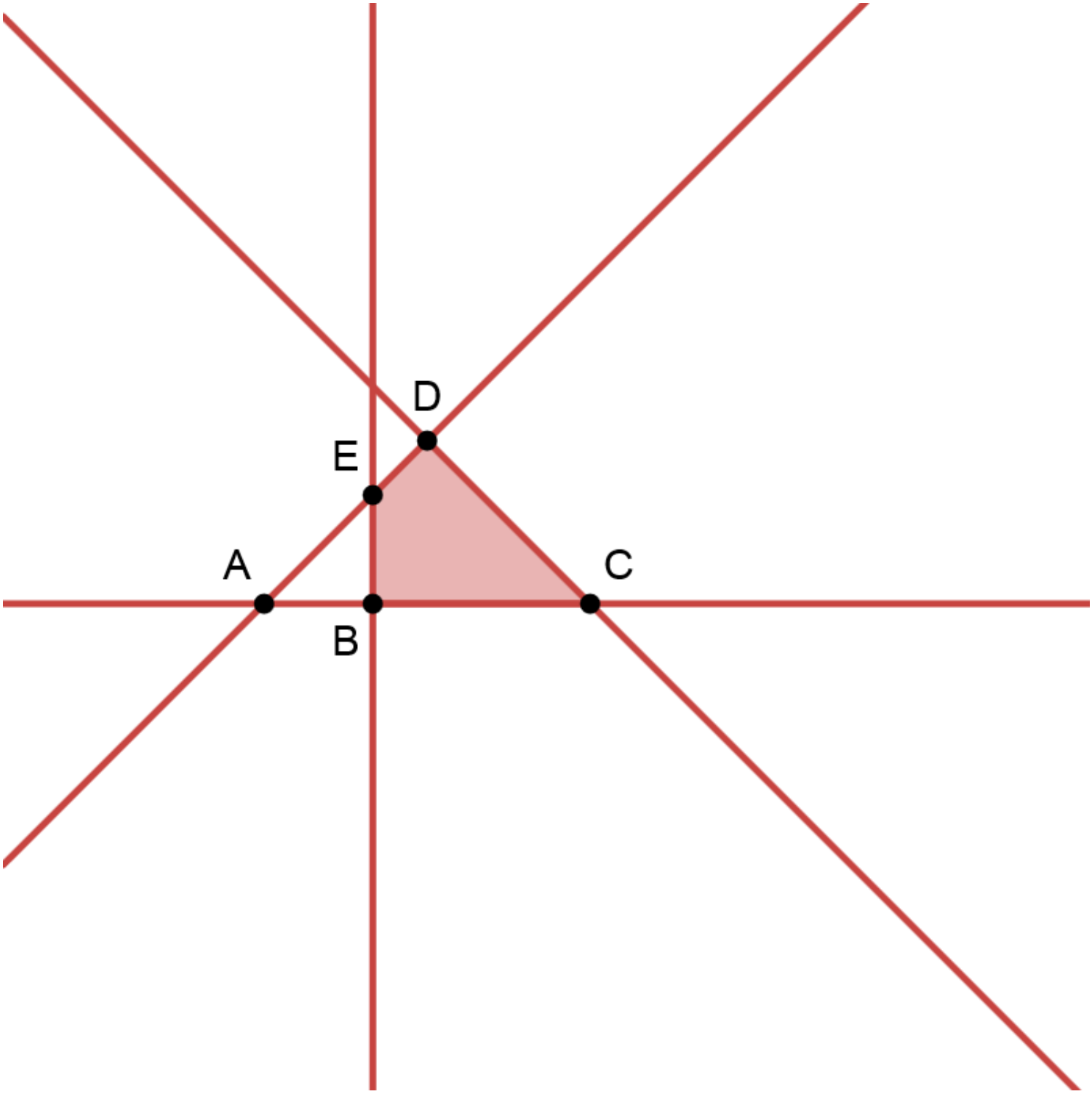
Lecture 8 (Some Polyhedron Properties)

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Adjacent Solutions

Two basic solutions are **adjacent** if they share $n-1$ LI constraints that are active. If two adjacent basic feasible solutions are both feasible, they are **adjacent BFS**.



If the above diagram, A and B are adjacent basic solutions. They are not adjacent BFS because A is not feasible. B and C are adjacent BFS.

Key Idea

The key takeaway for the simplex algorithm is this: We can more between adjacent BFS to more and more optimal solutions. When we cannot move anymore, then we are at the optimal BFS.

Standard Form

A polyhedron is in standard form if it is written in the form

$$P = \left\{ \vec{x} \in \mathbb{R}^n | A\vec{x} = \vec{b}, \vec{x} \geq 0 \right\}$$

for some $A \in \mathbb{R}^{m \times n}$, $\vec{b} \in \mathbb{R}^m$ and for rank $(A) = m$ (i.e. the polyhedron is full row rank). If we have m active constraints already from the matrix equation, we need to pick $n - m$ of the sign constraints to be binding in order to get a BFS.

Thm

$\vec{x}^* \in \mathbb{R}^n$ is a basic solution of P in standard form if and only if $A\vec{x}^* = \vec{b}$ and there exists some function $B : \{1, 2, \ldots, m\} \to \{1, 2, \ldots, n\}$such that

a) The columns $A_{B(1)}, A_{B(2)}, \ldots, A_{B(m)}$ are LI

b) If $i \neq B(1), B(2), \ldots, B(m)$ then $x_i^* = 0$

Proof

First, suppose $\vec{x} \in \mathbb{R}^n$ and suppose there exists B that satisfies **a)** and **b)**. Then the active constraints $x_i = 0$ for $i \neq B(1), B(2), \ldots, B(m)$ and $A\vec{x} = \vec{b}$ imply

$$\sum_{i=1}^{m} A_{B(i)} \vec{x}_{B(i)} = \sum_{i=1}^{n} A_i \vec{x}_i = A\vec{x} = \vec{b}$$

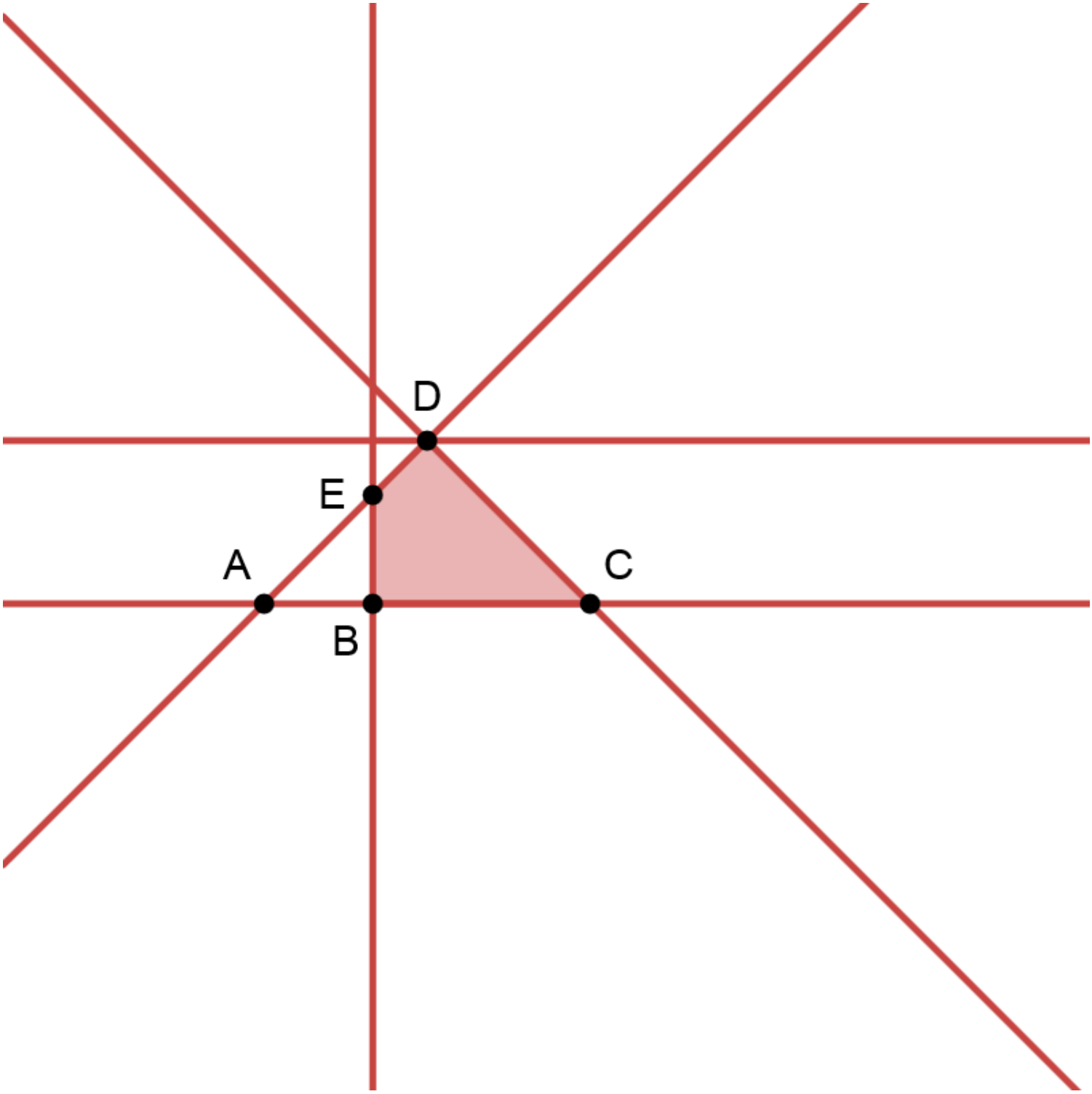
but as all of the $A_{B(i)}$ are LI (by **a)**), we see that there is a unique solution. As this solution has n LI constraints (due to the $\vec{x}_i = 0$ active constraints), then $\vec{x}$ is a basic solution. Now let's go the other way. Suppose $\vec{x}$ is a basic solution. Let $x_{B(1)}, x_{B(2)}, \ldots, x_{b(k)}$ be non-zero components of $\vec{x}$ (we cannot assume $k = m$). Then as $\vec{x}$ is a basic solution, we see

$$\sum_{i=1}^{n} A_i \vec{x}_i = \vec{b}$$

with $x_i = 0$ for all $i \neq B(1), \ldots, B(k)$ has a unique solution. As such, we know $A_{B(1)}, A_{B(2)}, \ldots, A_{B(k)}$ are LI. As such, we see $k \leq m$. Since we are assuming A has m LI rows, then we can pick $m - k$ of the remaining rows to form the remaining $m - k$ indices. This implies **a)** and **b)** will be satisfied.

Degenerate Solutions

A solution to an LP is **degenerate** if it has more than n active constraints.



In the above picture, A is a degenerate solution as there are 3 constraints active at A. In standard form, this is equivalent to the statement that more than $n - m$ of the x_j terms are equal to 0.

Polytopes

A bounded polyhedron is called a **polytope**. We will state two theorems about polytopes below. The proofs aren't going to be required for our purposes, but I will provide them just in case any students want to see them. These closely follow the proofs from Bertsimas and Tsitsiklis.

Thm

A non-empty polytope is the convex hull of its extreme points.

Proof

We previously showed every polyhedron is a convex set. This implies all convex combinations of the extreme points must remain in the polyhedron. What we need to show is that all points in the polytope, P, are a convex combination of the extreme points. Let $\dim(P)$ be the smallest integer, k, such that P is contained in a k dimensional affine subspace of $\mathbb{R}^n$. Now, apply induction on the dimension of P. First, for the base case, $k = 0$. A 0 dimensional polytope P is a single point. A single point will be an extreme point, so this statement holds for a 0 dimensional polytope. Now, suppose the induction hypothesis is true for all P such that $0 \leq \dim(P) < k$. If P is a k dimensional polytope, then P is contained in a k-dimensional affine subspace, S of $\mathbb{R}^n$. Suppose S is of the form:

$$S = \{\vec{x}_0 + \lambda_1\vec{x}_1 + \cdots + \lambda_k\vec{x}_k | \lambda_1, \ldots, \lambda_k \in \mathbb{R}\}$$

where each $\vec{x}_i \in \mathbb{R}^n$. Let $\vec{f}_1, \ldots \vec{f}_{n-k} \in \mathbb{R}^n$ be L.I. and orthogonal to each $\vec{x}_i$. Let $g_i = \vec{f}_i \cdot \vec{x}_0$ for each $i = 1, 2, \ldots, n-k$. Then each element $\vec{x} \in S$ has

$$\vec{f}_i \cdot \vec{x} = \vec{f}_i \cdot (\vec{x}_0 + \lambda_1\vec{x}_1 + \cdots + \lambda_k\vec{x}_k) = \vec{f}_i \cdot \vec{x}_0 + \lambda_1\vec{f}_i \cdot \vec{x}_1 + \cdots + \lambda_k\vec{f}_i\vec{x}_k = \vec{f}_i \cdot \vec{x}_0 = g_i$$

This also is true for each element of P as $P \subset S$. Let $\vec{z} \in P$. If $\vec{z}$ is an extreme point, then it is certainly in the convex hull of the extreme points of P (trivially). Suppose $\vec{z}$ is not an extreme point. Then we can choose an extreme point $\vec{y} \in P$ and take the points of the form: $\vec{u} = \vec{z} + \lambda(\vec{z} - \vec{y})$. Since P is bounded, then for large enough λ, this a point of this form will leave P. Upon doing so, one of the inequality constraints will be broken. Suppose this is the constraint is

$$\vec{a}_i^* \cdot \vec{x} \geq b_i^*$$

Then we can find some $\lambda^* > 0$ such that for

$$\vec{u} = \vec{z} + \lambda^*(\vec{z} - \vec{y})$$

we have equality

$$\vec{a}_i^* \cdot \vec{u} = b_i^*$$

Since the inequality constraint will break if $\lambda > \lambda^*$, it follows that

$$\vec{a}_i^* \cdot (\vec{z} - \vec{y}) < 0$$

(Note that both $\vec{z}, \vec{y}$ satisfy the inequality constraint. As we move from $\vec{z}$ towards $\vec{y}$ we get closer to breaking the inequality constraint, hence the < 0). Now define

$$Q = \{\vec{x} \in P | \vec{a}_i \cdot \vec{x} = b_i^*\}$$

As we saw $\vec{f}_i \cdot \vec{z} = \vec{f}_i \cdot \vec{y} = g_i$ we know $\vec{z} - \vec{y}$ is orthogonal to each $\vec{f}_i$. But we also know $\vec{a}_i^* \cdot (\vec{z} - \vec{y}) < 0$ so $\vec{a}_i^*$ is linearly independent from the vectors $\vec{f}_i$. As such, we see

$$Q \subset \left\{\vec{x} \in \mathbb{R}^n | \vec{a}_i^* \cdot \vec{x} = b_i^*, \vec{f}_i \cdot \vec{x} g_i\right\}$$

So Q is a subset of a subspace which is at most a $k-1$ dimensional. By the induction hypothesis on Q, we see that each $\vec{u}$ can be expressed in the form

$$\vec{u} = \sum_i \lambda_i \vec{v}_i$$

Where each $\vec{v}_i$ is an extreme point of Q. Each extreme point of Q will also be an extreme point of P. From λ^*, we see

$$\vec{z} = \frac{\vec{u} + \lambda^*\vec{y}}{1 + \lambda^*}$$

So

$$\vec{z} = \frac{\lambda^*\vec{y}}{1 + \lambda^*} + \sum_i \frac{\lambda_i}{1 + \lambda^*}\vec{v}_i$$

and thus $\vec{z}$ is a convex combination of the extreme points of P and therefore each point in P can be written as a convex combination of the extreme points of P.

Thm

Any linear transformation of a polyhedron (polytope) is a polyhedron (polytope).

Proof

We will apply projections. Suppose $\vec{x} \in \mathbb{R}^n$. Then if $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ then for $k \leq n$, the projection

$$\pi_k(\vec{x}) = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix}$$

projects $\vec{x}$ to its first k coordinates. For a whole set, we have

$$\Pi_k(S) = \{\pi_k(\vec{x}) \,|\vec{x} \in S\}$$

Notice this set is empty if and only if S is empty. Now to the proof. Suppose

$$P = \left\{ \vec{x} \in \mathbb{R}^n | \sum_j a_{ij} x_j \geq b_i,\ i = 1, 2, \ldots, m \right\}$$

Then we can get the projection $\Pi_{n-1}(P)$ by taking the following steps:
1) Rewrite each constraint into the form

$$a_{in} x_n \geq -\sum_{j=1}^{n-1} a_{ij} x_j + b_i$$

and if $a_{in} \neq 0$, then divide both sides by a_{in}. Now define $\vec{\bar{x}} = (x_1, \ldots x_{n-1})$. Then we get the equivalent form of the constraints:

$$\begin{cases} x_n \geq \vec{f_i} \cdot \vec{\bar{x}} & \text{if } a_{in} > 0 \\ d_j + \vec{f_i} \cdot \vec{\bar{x}} \geq x_n & \text{if } a_{jn} < 0 \\ 0 \geq d_k + \vec{f_i} \cdot \vec{\bar{x}} & \text{if } a_{kn} = 0 \end{cases}$$

for some scalars d_i, d_j, d_k and some vectors $\vec{f_i}, \vec{f_j}, \vec{f_k} \in \mathbb{R}^{n-1}$
2) Let Q be the polyhedron in $\mathbb{R}^{n-1}$ defined by

$$\begin{cases} d_j + \vec{f_j} \cdot \vec{\bar{x}} \geq d_i + \vec{f_i} \cdot \vec{\bar{x}} & \text{if } a_{in} > 0 \text{ and } a_{jn} < 0 \\ 0 \geq d_k + \vec{f_i} \cdot \vec{\bar{x}} & \text{if } a_{kn} = 0 \end{cases}$$

Notice this process reduces the dimension, but may add a large number of constraints. This process is called Fourier-Motzkin Elimination. Now, onto the theorem. We will prove a more general result, and will get our desired theorem as a result.

Lemma

The polyhedron Q built from the elimination algorithm is equal to the projection $\Pi_{n-1}(P)$ of P.

Proof

If $\vec{\bar{x}} \in \Pi_{n-1}(P)$, then $\exists x_n$ such that $\begin{pmatrix} \vec{\bar{x}} \\ x_n \end{pmatrix} \in P$. Notice that $\vec{x}$ satisfies the elimination step **1)** constraints and the step **2)** constraints (as a result of satisfying the step **1)** constraints). As such, we see $\Pi_{n-1}(P) \subseteq Q$. Now, suppose $\vec{\bar{x}} \in Q$. Then as the constraint

$$d_j + \vec{f_j} \cdot \vec{\bar{x}} \geq d_i + \vec{f_i} \cdot \vec{\bar{x}} \text{if } a_{in} > 0 \text{ and } a_{jn} < 0$$

holds for Q, then it follows that

$$\min_j d_j + \vec{f_j} \cdot \vec{x} \geq \max_i d_i + \vec{f_i} \cdot \vec{x}$$

for $a_{jn} < 0$, $a_{in} > 0$. Then any x_n between these two bounds above, we have

$$\begin{pmatrix} \vec{x} \\ x_n \end{pmatrix}$$

satisfying the three inequalities from step **1)**, thus $\begin{pmatrix} \vec{x} \\ x_n \end{pmatrix} \in P$. If we take repeated projections of polyhedra, we will continue to get polyhedra. Stated in another form:

Let $P \subseteq \mathbb{R}^{n+k}$ be a polyhedron. Then

$$Q = \left\{ \vec{x} \in \mathbb{R}^n | \exists \vec{y} \in \mathbb{R}^k \text{ such that } \begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} \in P \right\}$$

is also a polyhedron. Now doe the original theorem. We wish to show that if $P \subseteq \mathbb{R}^n$ is a polyhedron and $A \in \mathbb{R}^{m \times n}$, then

$$Q = \{ A\vec{x} | \vec{x} \in P \}$$

is also a polyhedron. Suppose

$$Q = \{ \vec{y} \in \mathbb{R}^m | \exists \vec{x} \in \mathbb{R}^n \text{ s.t. } A\vec{x} = \vec{y},\ \vec{x} \in P \}$$

Then Q is the projection of

$$\left\{ \begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} \in \mathbb{R}^{n+m} | A\vec{x} = \vec{y}, \vec{x} \in P \right\}$$